

Dhafer Malouche, Ph.D.

Professor of Statistics

Department of Mathematics, Statistics, and Physics, College of Arts and Sciences
Qatar University, Doha, Qatar

dhafer.malouche@qu.edu.qa · dhafermalouche.net · GitHub: [malouche](#) · Google Scholar · ORCID
0000-0002-0494-7141

Last updated: September 2026

RESEARCH INTERESTS

Bayesian networks and graphical models, Biostatistics and health outcomes research, Qatar Biobank data analysis, Path analysis and causal inference, Meta-analysis and systematic reviews (Bayesian and frequentist evidence synthesis), Chronic pain and mental health modeling, Survey methodology, Quantitative analysis of regulatory frameworks, Electoral data, Educational statistics, Statistical computing and data visualization, Consumer preferences and sensory data analysis.

Citation metrics (Google Scholar, September 2026): 997 citations, *h*-index 16, *i10*-index 27 (since 2021: 463 citations, *h*-index 13, *i10*-index 17).

EMPLOYMENT HISTORY

- Aug. 2022 – present** **Professor of Statistics**, Qatar University, Doha, Qatar.
Department of Mathematics, Statistics, and Physics, College of Arts and Sciences.
Teaching (2022–present): Statistics II, Linear Algebra, Multivariate Analysis, Stochastic Processes, Actuarial Statistics I&II, Bayesian Statistics, Statistical Computation and Simulation, Bayesian Statistical Inference, Advanced Biostatistics (for Ph.D. candidates), Experimental Design and Statistical Analysis, Senior Project, Critical Thinking (Honors Program), Expertise, Experience and Exchange: tips for research methods (Honors Program). See also *Teaching* below.
- Jul. 2021 – Jul. 2022** **Professor of Statistics**, The American University in Cairo, Cairo, Egypt.
Department of Mathematics and Actuarial Science, School of Sciences and Engineering.
Teaching: Statistical Inference, Introduction to Statistics, Data Science with Python.
- 2003 – 2021** **Professor / Associate Professor / Assistant Professor**, University of Carthage, Tunis, Tunisia.
École Supérieure de la Statistique et de l'Analyse de l'Information (ESSAI).
Positions: Professor (from 2018), Associate Professor (2011–2018), Assistant Professor (2003–2011).
Teaching: Data Analysis (PCA, CA, MCA) with R; Mathematical Statistics with R; Data Mining with R; Bayesian Statistics with OpenBUGS/WinBUGS/RStan/JAGS; Time Series (ARMA, SARIMA) with R/Python; Big Data with R/Python, SQL, Spark, H2O; Advanced R/Python: data management, data visualization, dashboards, Shiny apps, Heroku apps, Bokeh.
Research: supervised 5 Ph.D. theses (swallowing-sound detection and classification, sensory analysis, genetics, zoonotic cutaneous leishmaniasis incidence, cardiovascular risk factors) and many Master and Engineering theses.
Administration: Director of the Department of Statistics (2004–2007).
- 2014 and 2019** **Visiting Associate Professor / Associate Research Scholar**, Yale University, New Haven, USA.
Whitney and Betty MacMillan Center for International and Area Studies and Department of Statistics and Data Science.
Positions: Visiting Associate Professor (2014), consulting on several projects with Yale scholars (2015–2018), Associate Research Scholar (2019).
Missions: Democratic Transition in Tunisia, Governance and Local Development (Program on Governance and Local Development, GLD, now at the University of Gothenburg): two face-to-face surveys in Tunisia; G-econ team on building local GDP data; teaching Time Series with R/Python.

- 2016 – 2017 Visiting Fulbright Scholar**, University of Michigan, Ann Arbor, USA.
Center for Political Studies, Institute for Social Research. Research on data quality, survey methodology and interviewer effects. Teaching: 4 lectures on applied graphical models, 1 on data visualization with R, 1 on sensory analysis.
- 2011 Visiting Fulbright Scholar**, Stanford University, Palo Alto, USA.
Department of Statistics. Research on graphical models, the faithfulness assumption, and covariance graphs.
- 2002 – 2003 Visiting Assistant Professor**, York University, Toronto, Canada.
Department of Statistics. Teaching: Applied Regression Models with SAS, Introduction to the Theory of Probability, Introduction to Statistics with Minitab. Research: Monte Carlo methods and Bayesian estimation of graphical models.
- 1998 – 2002 Assistant Professor**, University of Sousse, Sousse, Tunisia.
Institut Préparatoire aux Écoles d'Ingénieurs. Teaching: Analysis, Calculus, Algebra. Research: natural exponential families, Pick functions, Markov chains. Administration: Director of the Department of Mathematics.

Consulting and other appointments

- WHO Tunisia, senior data analyst consultant (05/2020–05/2021): impact of COVID-19 on psychiatric needs, hotline data.
- WARC Africa, Sierra Leone, data analyst consultant (05/2020–01/2021): online daily monitoring platform for a survey.
- Covidradar24.org / Rosettahub.com, data analyst master contributor (10/2020–12/2022).
- Fellow, Pan-African Scientific Research Council (08/2020–08/2022).
- World Bank / Program on Governance and Local Development: co-developer of the [Local Governance Performance Index \(LGPI\)](#); pilot survey in Tunisia, February 2015 (World Bank document 106241, May 2016).

EDUCATION

- Sep. 2009 Habilitation (Tenure), Statistics**, Université de Tunis El Manar, Tunisia. École Nationale d'Ingénieurs de Tunis.
Thesis: *Problèmes autour de la probabilité et de la statistique : méthodes et applications*.
Dissertation: malouche.github.io/myCV/reports.html
- Oct. 1997 Doctorate (Ph.D.), Statistics**, Paul Sabatier University, Toulouse, France.
Thesis: *Classification des familles exponentielles associées à des fonctions Pick*. Dissertation: malouche.github.io/myCV/reports.html
- 1993 – 1994 Master's Degree (D.E.A.), Statistics**, Paul Sabatier University, Toulouse, France. Applied Mathematics, Statistics.
- 1989 – 1993 Bachelor (Maîtrise)**, École Normale Supérieure de Bizerte, Tunisia.

TEACHING

All statistics courses are taught with R. The last column gives the programming and data-analysis content of each course.

Qatar University (2022 – present)

Code	Title	Level	Terms	Programming / data analysis
STAT 102	Statistics II	Undergraduate	Fall 2022; Fall 2023 (lecture and lab sections)	R (introduced to students in this course)
STAT 312	Stochastic Processes	Undergraduate	Fall 2022; Fall 2023; Fall 2024; Fall 2025 (2 sections); Fall 2026	R and RStudio throughout; R labs on Markov chains (<code>markovchain</code>), n -step transitions, simulation and visualisation, classification and absorption; since 2025, GitHub for code and submissions and GitHub Copilot in RStudio under written guidelines on responsible AI use

Code	Title	Level	Terms	Programming / data analysis
STAT 341	Actuarial Statistics I	Undergraduate	Fall 2023; Fall 2024; Fall 2025 (2 sections)	R; computing methods for aggregate claims distributions
STAT 442	Actuarial Statistics II	Undergraduate	Spring 2023; Spring 2025	R for data analysis
STAT 481	Multivariate Analysis	Undergraduate	Fall 2026	All computational work in R with RStudio; five R labs (matrix algebra, normality assessment, Hotelling's T^2 , MANOVA, PCA, factor analysis); lab reports and course project submitted as reproducible Quarto documents
STAT 482	Bayesian Statistics	Undergraduate	Spring 2026	Weekly R labs (Beta–Binomial updating, grid approximation, posterior sampling, Bayesian regression); R, Stan and WinBUGS
STAT 499	Senior Project	Undergraduate	Spring 2025; Spring 2026	Supervised projects; Spring 2026 projects built in R on NHANES data
STAT 617	Statistical Computation and Simulation	Master	Fall 2023	Advanced R programming; Monte Carlo methods, random variate generation, bootstrap, MCMC and optimisation in R; term project RStudio
STAT 626	Bayesian Statistical Inference	Master	Spring 2023; Spring 2025	Supervised thesis
STAT 690	Thesis	Master	Fall 2025	R for advanced statistical analysis; in 2025, GitHub submissions and GitHub Copilot in RStudio under written guidelines
BIOL 601	Advanced Biostatistics	Doctorate (Ph.D.)	Fall 2024; Fall 2025	R (co-taught course); GitHub submissions and GitHub Copilot in RStudio under written guidelines
BIOL 503	Experimental Design and Statistical Analysis	Master	Spring 2025	—
MATH 231	Linear Algebra	Undergraduate	Spring 2024	R
HONS 301	Expertise, Experience and Exchange: tips for research methods (Honors Program)	Undergraduate	Fall 2024; Spring 2025; Fall 2025	—
HONS 203	Mastering Critical Thinking (Honors Program)	Undergraduate	Fall 2023 (2 sections)	—

Earlier institutions

- **The American University in Cairo (2021–2022):** Data Science with Python; Statistical Inference; Introduction to Statistics.
- **University of Carthage, ESSAI (2003–2021):** Data Analysis (PCA, CA, MCA) with R; Mathematical Statistics with R; Data Mining with R; Bayesian Statistics with OpenBUGS/WinBUGS/RStan/JAGS; Time Series with R/Python; Big Data with R/Python, SQL, Spark, H2O; Advanced R/Python: data management, data visualization, dashboards, Shiny apps, Heroku apps, Bokeh.
- **Yale University (2014–2019 period):** Time Series with R/Python.
- **University of Michigan (2016–2017):** 4 lectures on applied graphical models; 1 lecture on data visualization with R; 1 lecture on sensory analysis.
- **York University (2002–2003):** Applied Regression Models with SAS; Introduction to Statistics with Minitab; Introduction to the Theory of Probability.

SUPERVISION

Ph.D. theses (University of Carthage; all completed)

- 2019-06-28** Safa AOUINTI. Statistical treatment of NGS results from IMGT/HighV-QUEST of antigen receptors (immunoglobulins and T receptors): methodology and visualization.
- 2019-06-22** Khoulood TALMOUDI JABBARI. Spatio-temporal modeling of zoonotic cutaneous leishmaniasis (ZCL) in central Tunisia.
- 2019-05-21** Hajer KHLAIFI. Preliminary study on the detection and classification of swallowing sounds.
- 2019-02-14** Olfa SAIDI. Modeling the Tunisian burden of cardiovascular diseases and diabetes.
- 2018-07-02** Ibtihel REBHI. Statistical methods for sensory and consumer data mapping.

M.Sc. theses (Qatar University)

- 2026 – 2027** Hadeel ALASHWAL, M.Sc. Applied Statistics (thesis option). Control-spectrum shift, statistical calibration, and performance portability in routine biomarker prediction of liver steatosis and fibrosis (NHANES and Qatar primary-care data). Supervisor (co-supervisor: A. Al-Khinji). In progress.
- 2025 – 2027** Muna Abdalla RAYASHI, M.Sc. Statistics (thesis option). Comprehensive health phenotyping and predictive risk stratification: an integrated machine learning approach to gender disparities in the Qatari population (Qatar Biobank). Supervisor. In progress.

Undergraduate capstone and senior projects (Qatar University)

- Spring 2026** Temporal machine learning classification of anemia status using iron biomarkers: a 20-year NHANES analysis (2001–2020). Supervisor.
- Spring 2026** Unsupervised learning and autoencoders for iron deficiency phenotyping across the lifespan: temporal patterns in NHANES (2001–2020). Supervisor.
- Spring 2026** Deep learning regression for predicting serum ferritin from demographic and dietary factors using NHANES data (2017–2018 cycle). Supervisor (STAT 499 Senior Project).
- Fall 2025** Bibliometric study of CAR T-cell therapy in lymphoma (three students). Supervisor, with A. Al-Khinji. Presented 17 December 2025.
- Spring 2025** From regional to global: Qatar University’s research trajectory, 2010–2020 — a bibliometric study of ten MENA and global universities (STAT 499 Senior Project). Presented May 2025.
- Fall 2024** Autoimmune disease in the COVID-19 era: a literature-based synthesis with a machine-learning analysis plan (logistic regression, support vector machines, random forests) for diet and other predictors of autoimmune status (two students). Linked to QBB project QF-QBB-RES-ACC-00277; no Qatar Biobank data were used. Presented December 2024.

Other supervision

- University of Carthage: Master and Engineering theses on applied statistics and data science.
- Qatar University: supervisor of the two QU student research grants listed under Grants.

GRANTS

- 2024 – 2026** **Autoimmune disorders profile pre and post-COVID era**, QF-QBB-RES-ACC-00277. Prevalence and management of autoimmune diseases in Qatar; prognostic models to inform public health policy, based on Qatar Biobank data from 2015–2023.
- 2024 – present** **Investigating Gender Disparities in Quality of Life Across the Qatari Population**, QF-QBB-RES-ACC-00281. Health-related quality of life (HRQOL) disparities in Qatar, focusing on gender differences, using Qatar Biobank data and machine learning.
- 2025 – 2026** **Proteomic Signatures of Fatigue and Mental-Health Symptoms in the Qatar Biobank Cohort**, QF-QBB-RES-ACC-00374. Proteomic profiling to identify inflammatory and metabolic signatures of chronic fatigue, integrated into Bayesian networks to test pain–fatigue–mental health pathways and sex-specific profiles.
- Fall 2024** **Detecting the Undetectable: A Study on AI Tool Efficacy in Academic Writing**, QU Student Grant 2024 Cycle 2, no. 130. Efficacy of AI-detection tools in identifying AI-generated text in academic settings.
- Spring 2025** **Comparative Analysis of Research Publication Trends**, QU Student Grant 2025 Cycle 1, no. 293. Research output, citation impact and collaboration networks of leading MENA and global universities (2010–2020).

SKILLS

Languages English, French, and Arabic (strong reading, writing and speaking).

Coding R (RStudio, Quarto, Shiny, Stan), Python, SQL, Spark, H2O, Tableau, Git/GitHub, L^AT_EX.

Misc. Quantitative and qualitative research, survey design, project management, data mining, machine learning, big data, teaching, training, consultation, L^AT_EX typesetting and publishing.

COMMENTARY AND PUBLIC WRITING

- Benstead, L., Lust, E., Malouche, D., & Wichmann, J. (2014, 27 Oct). [Tunisian elections bring hope in uncertain times](#). *Washington Post*, Monkey Cage.
- Benstead, L., Lust, E., Malouche, D., Soltan, G., & Wichmann, J. (2013, 14 Feb). [Islamists aren't the obstacle](#). *Foreign Affairs*.
- Benstead, L. J., Lust, E. M., Malouche, D., Soltan, G., & Wichmann, J. (2013). [Winners and losers after Arab Spring](#). *YaleGlobal Online*.
- Benstead, L. J., Lust, E., & Malouche, D. (2012, 19 Dec). [Tunisian revolution is work in progress](#). *YaleGlobal Online*.
- Malouche, D. (2019). [Arab Spring success story: Tunisians vote](#). *YaleGlobal Online*.
- Malouche, D. (2020, Oct). [Quel impact de la Covid-19 sur l'espérance de vie en Tunisie ?](#) *Leaders* (Tunisia).
- Malouche, D. (2021, Jul). Un travail d'analyse statistique exige la disponibilité des données du ministère de la santé. *Espace Manager* (Tunisia).

PUBLICATIONS

Accepted papers are listed first. Published journal articles are grouped by theme and listed newest first; conference proceedings and the early statistical-methodology papers follow as their own groups. Quartile tags are given as published. Preprints are listed on dhafermalouche.net.

Accepted (in press)

1. Muhammad, M. I., Al-Khinji, A., Al-Ansari, M., Al-Marri, N., Alshaheen, S., & Malouche, D. (2026). Allele-stratified activity of RAS inhibitors in pancreatic ductal adenocarcinoma: a systematic review with model-validation-stratified synthesis of preclinical evidence and exploratory meta-analysis of early-phase trials. *Cancers*. [Accepted, in press]
2. Mzoughi, C., & Malouche, D. (2026). Legislating the future: statistical foresight as a foundation for anticipatory law. Conference paper, [12th Annual International Academic Conference](#), Kuwait International Law School (Kuwait, 6–7 May 2026); to appear in the conference supplement of the *Kuwait International Law School Journal* (peer-reviewed). [Full paper accepted 9 July 2026]

Published

Bayesian networks, path analysis and health outcomes (Qatar Biobank, Qatar primary care, clinical cohorts)

3. Al-Khinji, A., Malouche, D., Al-Hor, A., Al Tamimi, G., Al-Korbi, N., Al-Kuwari, H., Al Rewaily, D., Al-Matwi, M., Haj Bakri, A., & Al-Kuwari, M. G. (2026). [Autoimmune disorders and cardiovascular disease: a matched cross-sectional analysis of disease-specific associations in Qatari primary care](#). *American Heart Journal Plus: Cardiology Research and Practice*, 71, 100907. doi:10.1016/j.ahjo.2026.100907
4. Al-Sharman, A., Khalil, H., Malouche, D., Kanaan, S., Kim, M., Saad, N., & Abdelrazeq, M. (2026). [Directional symptom dependencies in multiple sclerosis and Parkinson's disease: a comparative Bayesian network analysis](#). *Journal of Multidisciplinary Healthcare*, 19, 1–19. doi:10.2147/JMDH.S607760
5. Al-Khinji, A., Malouche, D., Al-Hor, A., Haj Bakri, A., & Al-Kuwari, M. G. (2026). [Differential laboratory monitoring in autoimmune disease: a matched case–control study from Qatar primary care](#). *Journal of Translational Medicine*, published online 15 July 2026. [Q1] doi:10.1186/s12967-026-08576-2
6. Al-Khinji, A., & Malouche, D. (2026). [Probabilistic pathways from chronic pain to suicidal ideation in Qatar: a Bayesian-network study](#). *Journal of Translational Medicine*, 24, 899. [Q1] doi:10.1186/s12967-026-08286-9

7. Al-Khinji, A., Malouche, D., Al-Thani, N., Mustafa, A., Abdulmajeed, J., & Al-Kuwari, M. G. (2026). [Hematological abnormalities in clinically diagnosed non-alcoholic steatohepatitis: prevalence, clinical correlates, and fibrosis risk in a case-control study from Qatar](#). *Frontiers in Medicine*, 13, 1773499. [Q1] doi:10.3389/fmed.2026.1773499
8. Al-Khinji, A., Yusuf, M., Al Ashwal, H., Al Hor, A., Habib, A., Malouche, D., Al-Kuwari, M. G., & Haj Bakri, A. (2026). [Translating vaccine durability data into personalized protection for autoimmune disease populations](#). *Journal of Clinical and Translational Science*, 10(s1), 12–13 (congress abstract). doi:10.1017/cts.2026.10272
9. Al-Hor, A., Al-Khinji, A., Malouche, D., & Al-Kuwari, M. (2026). [Comparative cardiovascular risk burden across autoimmune diseases: evidence from a primary care cohort](#). *Journal of Clinical and Translational Science*, 10(s1), 160 (congress abstract). doi:10.1017/cts.2026.10636
10. Al-Khinji, A., Malouche, D. (2025c). [Modeling chronic pain interconnections using Bayesian networks: insights from the Qatar Biobank study](#). *Frontiers in Pain Research*, 6, 1573465. [Q1] doi:10.3389/fpain.2025.1573465
11. Gad, A., Malouche, D., Chhabra, M., Hoang, D., Suk, D., Ron, N., ... Elmakaty, I. (2024). [Impact of birth weight to placental weight ratio and other perinatal risk factors on left ventricular dimensions in newborns: a prospective cohort analysis](#). *Journal of Perinatal Medicine*, 52(4), 433–444. [Q2] doi:10.1515/jpm-2023-0384
12. Khalil, H., Malouche, D., Al Marri, D., Shoukry, M., Shehabi, M., Yahia, R., ... El-Salim, K. (2024). [The comparative ability of five mobility and balance outcome measures in predicting the risk of falls in people with multiple sclerosis](#). *Multiple Sclerosis and Related Disorders*, 92, 105971 (congress abstract). [Q2] doi:10.1016/j.msard.2024.105971
13. Khalil, H., Malouche, D., Kanaan, S., Al-Sharman, A., & El-Salim, K. (2024). [Factors affecting the quality of sleep and physical activity participation of multiple sclerosis \(MS\) patients: a path analysis](#). *Multiple Sclerosis and Related Disorders*, 92, 106099 (congress abstract). [Q2] doi:10.1016/j.msard.2024.106099

Meta-analysis and systematic reviews

14. Al-Khinji, A., Alashwal, H., Al-Hor, A., Al-Kuwari, H., Bismi, F., Maki, N. A., Abunada, T., & Malouche, D. (2026). [Bayesian and frequentist meta-analysis of GCB-classification heterogeneity in TP53-reporting DLBCL cohorts at small k: an applied case study with reporting guidance](#). *BMC Medical Research Methodology*, published online 21 September 2026. doi:10.1186/s12874-026-03015-8
15. Al-Khinji, A., Yusuf, M., Al-Thani, N., Al Ashwal, H., Al Hor, A., Al-Kuwari, H., Habib, A., & Malouche, D. (2026). [Why TP53 expression varies in lymphoma: insights from a Bayesian meta-analysis](#). *Journal of Clinical and Translational Science*, 10(s1), 23 (congress abstract). doi:10.1017/cts.2026.10297
16. Al Kuwari, H., Al Khinji, A., Al-Hamed, D., Al Hor, A., Malouche, D., & Iqbal, J. (2025). [NOTCH1 mutation status as a prognostic biomarker in T-cell acute lymphoblastic leukemia: a systematic review and meta-analysis](#). *BMC Cancer*, 25(1), 1820. [Q2] doi:10.1186/s12885-025-15233-2
17. Al-Khinji, A., Al-Korbi, N., Al-Kuwari, S., Al-Hor, A., & Malouche, D. (2025b). [Immune checkpoint inhibitors in pancreatic adenocarcinoma: a systematic review and meta analysis of clinical outcomes](#). *Frontiers in Oncology*, 15, 1569884. [Q2] doi:10.3389/fonc.2025.1569884

Education and learning technologies

18. Mizza, D., ElKhodary, K., Malouche, D., & Al-Mousa, M. (2026). [Multi-modal virtual reality in engineering education: a design case study of fostering situational interest through project-based learning](#). *International Journal of Engineering Pedagogy (iJEP)*, 16(5), 4–27. doi:10.3991/ijep.v16i5.61389
19. Mizza, D., Reese, M., & Malouche, D. (2025). [Flipped classroom evaluation and blended learning potential: a case study of engagement and inclusion in quantitative education](#). *Smart Learning Environments*, 12, 56. [Q1] doi:10.1186/s40561-025-00412-2

Survey research and governance

20. Benstead, L. J., Kao, K., Landry, P. F., Lust, E. M., & Malouche, D. (2017). [Using tablet computers to implement surveys in challenging environments](#). *Survey Practice*, 10(2), 1–9. doi:10.29115/SP-2017-0009

Population health, epidemiology and health policy (Tunisia)

21. Saidi, O., Malouche, D., Saksena, P., Arfaoui, L., Talmoudi, K., Hchaichi, A., ... Ben Alaya, N. (2021). [Impact of contact tracing, respect of isolation and lockdown in reducing the number of cases infected with COVID-19: case study: Tunisia's response from March 22 to 04 May 2020](#). *International Journal of Infectious Diseases*, 113, 26–33. [Q1] doi:10.1016/j.ijid.2021.02.010
22. Saidi, O., Hajjem, S., Zoghlami, N., Aounallah-Skhiri, H., Mansour, N. B., Hsairi, M., ... O'Flaherty, M. et al. (2019). [Premature mortality attributable to smoking among Tunisian men in 2009](#). *Tobacco Induced Diseases*, 17. [Q1] doi:10.18332/tid/112666

23. Saidi, O., O'Flaherty, M., Zoghalmi, N., Malouche, D., Capewell, S., Critchley, J. A., ... Guzman Castillo, M. (2019). [Comparing strategies to prevent stroke and ischemic heart disease in the Tunisian population: Markov modeling approach using a comprehensive sensitivity analysis algorithm](#). *Computational and Mathematical Methods in Medicine*, 2019, 2123079. [Q2] doi:10.1155/2019/2123079
24. Saidi, O., Zoghalmi, N., Bennett, K. E., Mosquera, P. A., Malouche, D., Capewell, S., ... O'Flaherty, M. (2019). [Explaining income-related inequalities in cardiovascular risk factors in Tunisian adults during the last decade: comparison of sensitivity analysis of logistic regression and Wagstaff decomposition analysis](#). *International Journal for Equity in Health*, 18(1), 177. [Q1] doi:10.1186/s12939-019-1047-6
25. Salem, S., Malouche, D., & Romdhane, H. B. (2019). [Tunisian population quality of life: a general analysis using SF-36](#). *Eastern Mediterranean Health Journal*, 25(9), 613–621. [Q3] doi:10.26719/emhj.18.030
26. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017a). [Comparative performance analysis for generalized additive and generalized linear modeling in epidemiology](#). *International Journal of Advanced Computer Science and Applications*, 8(12). [Q3] doi:10.14569/IJACSA.2017.081255
27. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017b). [Modeling zoonotic cutaneous leishmaniasis incidence in central Tunisia from 2009–2015: forecasting models using climate variables as predictors](#). *PLoS Neglected Tropical Diseases*, 11(8), e0005844. [Q1] doi:10.1371/journal.pntd.0005844
28. Kerfai, N., Bejar Ghadhab, B., & Malouche, D. (2016). [Performance measurement and quality costing in Tunisian manufacturing companies](#). *The TQM Journal*, 28(4), 588–596. [Q1] doi:10.1108/TQM-10-2013-0119
29. Saidi, O., Malouche, D., O'Flaherty, M., Mansour, N. B., Skhiri, H., Romdhane, H. B., & Bezdah, L. (2016). [Assessment of cardiovascular risk in Tunisia: applying the Framingham risk score to national survey data](#). *BMJ Open*, 6(11), e009195. [Q1] doi:10.1136/bmjopen-2015-009195
30. Saidi, O., O'Flaherty, M., Mansour, N. B., Aissi, W., Lassoued, O., Capewell, S., ... Romdhane, H. B. (2015). [Forecasting Tunisian type 2 diabetes prevalence to 2027: validation of a simple model](#). *BMC Public Health*, 15(1), 104. [Q1] doi:10.1186/s12889-015-1416-z
31. Karaoud, M., Bouafif, N., Malouche, D., Kouni, C., & Achour, N. (2014). [La mortalité parmi les enfants âgés de moins de 15 ans en Tunisie peut être liée à la température](#). *Revue d'Épidémiologie et de Santé Publique*, 62, S122 (congress abstract). [Q3] doi:10.1016/j.respe.2014.05.021
32. Karaoud, M., Malouche, D. et al. (2014). [Les effets de la température sur la mortalité chez les personnes âgées en Tunisie](#). *Revue d'Épidémiologie et de Santé Publique*, 62, S219–S220 (congress abstract). [Q3] doi:10.1016/j.respe.2014.06.155
33. Karaoud, M., Malouche, D., & Alaya, N. B. (2014). [Mortalité et température journalière en Tunisie : une étude multi-région](#). *Revue d'Épidémiologie et de Santé Publique*, 62, S218 (congress abstract). [Q3] doi:10.1016/j.respe.2014.06.149
34. Karaoud, M., Malouche, D., & Bouafif, N. (2014). [Méthodologie de l'analyse de la relation température–mortalité en Tunisie](#). *Revue d'Épidémiologie et de Santé Publique*, 62, S141–S142 (congress abstract). [Q3] doi:10.1016/j.respe.2014.05.066
35. Fehri, R., Rifi, H., Alboueiri, A., Malouche, D., Ayadi, M., Rais, H., & Mezlini, A. (2013). [Carcinoma of unknown primary: retrospective study of 437 patients treated at Salah Azaiez Institute](#). *La Tunisie Médicale*, 91(3), 205–208. [Q4]
36. Arfa, I., Nour, S., Abid, A., Alaya, N. B.-B., Zorgati, M., Malouche, D., ... Romdhane, H. B. et al. (2010). [Absence d'association entre les polymorphismes du système rénine angiotensine \(SRA\) et l'hypertension artérielle chez les diabétiques de type 2 tunisiens](#). *La Tunisie Médicale*, 88(01), 37–40. [Q4]
37. Arfa, I., Nour, S., Abid, A., Bouafif-Ben, N. A., Zorgati, M. M., Malouche, D., ... Ben, H. R. et al. (2010). [Lack of association between renin-angiotensin system \(RAS\) polymorphisms and hypertension in Tunisian type 2 diabetics](#). *La Tunisie Médicale*, 88(1), 38–41. [Q4]
38. Arfa, I., Abid, A., Nour, S., Elloumi-Zghal, H., Malouche, D., Mannai, I., ... Zouari, B. et al. (2008). [Lack of association between the angiotensin-converting enzyme gene \(I/D\) polymorphism and diabetic nephropathy in Tunisian type 2 diabetic patients](#). *Journal of the Renin-Angiotensin-Aldosterone System*, 9(1), 32–36. [Q2] doi:10.3317/jraas.2008.002
39. Arfa, I., Abid, A., Malouche, D., Alaya, N. B., Azegue, T. R., Mannai, I., ... Blousa-Chabchoub, S. et al. (2007). [Familial aggregation and excess maternal transmission of type 2 diabetes in Tunisia](#). *Postgraduate Medical Journal*, 83(979), 348–351. [Q1] doi:10.1136/pgmj.2006.053744

40. Aouinti, S., Giudicelli, V., Duroux, P., Malouche, D., Kossida, S., & Lefranc, M.-P. (2016). [IMGT/StatClonotype for pairwise evaluation and visualization of NGS IG and TR IMGT clonotype \(AA\) diversity or expression from IMGT/HighV-QUEST](#). *Frontiers in Immunology*, 7, 339. [Q1] doi:10.3389/fimmu.2016.00339
41. Aouinti, S., Malouche, D., Giudicelli, V., Kossida, S., & Lefranc, M.-P. (2016). [Correction: IMGT/HighV-QUEST statistical significance of IMGT clonotype \(AA\) diversity per gene for standardized comparisons of next generation sequencing immunoprofiles of immunoglobulins and T cell receptors](#). *PLoS ONE*, 11(1), e0146702. doi:10.1371/journal.pone.0146702
42. Aouinti, S., Malouche, D., Giudicelli, V., Kossida, S., & Lefranc, M.-P. (2015). [IMGT/HighV-QUEST statistical significance of IMGT clonotype \(AA\) diversity per gene for standardized comparisons of next generation sequencing immunoprofiles of immunoglobulins and T cell receptors](#). *PLoS ONE*, 10(11), e0142353. [Q1] doi:10.1371/journal.pone.0142353
43. Ghouila, A., Jmel, H., Yahia, S., Malouche, D., & Abdelhak, S. (2009). Multi-SOM: a novel clustering approach for gene expression analysis. *Infection, Genetics and Evolution*, 9(3), 374 (congress abstract). [Q1]
44. Ghouila, A., Yahia, S. B., Malouche, D., Jmel, H., Laouini, D., Guerfali, F. Z., & Abdelhak, S. (2009). [Application of Multi-SOM clustering approach to macrophage gene expression analysis](#). *Infection, Genetics and Evolution*, 9(3), 328–336. [Q1] doi:10.1016/j.meegid.2008.09.009

Sensory and consumer science; statistical software

45. Rebhi, I., & Malouche, D. (2023). [SensMap R package and SensMapGUI Shiny web application for sensory and consumer data mapping: variations on external preference mapping and stability assessment](#). *Journal of Sensory Studies*, 38(2), e12809. [Q3] doi:10.1111/joss.12809
46. Ben-Hassine, K., Taamalli, A., Rezig, L., Yanguì, I., Benincasa, C., Malouche, D., ... Mnif, W. (2022). [Effect of processing technology on chemical, sensory, and consumers' hedonic rating of seven olive oil varieties](#). *Food Science & Nutrition*, 10(3), 863–878. [Q1] doi:10.1002/fsn3.2717
47. Ben-Hassine, K., Yanguì, I., Mnif, W., Taamalli, A., Benincasa, C., Kamoun, N., & Malouche, D. (2022). [Chemometric analysis and physicochemical composition of foreign and Tunisian olive oil: consumer preferences](#). *Journal of Food Quality*, 2022, 3981028. [Q2] doi:10.1155/2022/3981028
48. Kongbonga, G. Y. M., Hassine, K. B., Ghalila, H., Malouche, D. et al. (2019). [Front-face fluorescence using UV-LED coupled to USB spectrometer to discriminate between virgin olive oil from two cultivars](#). *Food and Nutrition Sciences*, 10(2), 119–127. [Q2] doi:10.4236/fns.2019.102010
49. Rebhi, I., & Malouche, D. (2017a). [An approach for external preference mapping improvement by denoising consumer rating data](#). *International Journal of Advanced Computer Science and Applications*, 8(12), 500–508. [Q3] doi:10.14569/IJACSA.2017.081266
50. Hassine, K. B., Taamalli, A., Slama, M. B., Khouloud, T., Kiristakis, A., Benincasa, C., ... Bornaz, S. et al. (2015). [Characterization and preference mapping of autochthonous and introduced olive oil cultivars in Tunisia](#). *European Journal of Lipid Science and Technology*, 117(1), 112–121. [Q2] doi:10.1002/ejlt.201400049
51. Hassine, K. B., El Riachy, M., Taamalli, A., Malouche, D., Ayadi, M., Talmoudi, K., ... Romano, E. et al. (2014). [Consumer discrimination of Chemlali and Arbequina olive oil cultivars according to their cultivar, geographical origin, and processing system](#). *European Journal of Lipid Science and Technology*, 116(7), 812–824. [Q2] doi:10.1002/ejlt.201300254
52. Ben-Hassine, K., Taamalli, A., Ferchichi, S., Mlaouah, A., Benincasa, C., Romano, E., ... Hammami, M. (2013). [Physicochemical and sensory characteristics of virgin olive oils in relation to cultivar, extraction system and storage conditions](#). *Food Research International*, 54(2), 1915–1925. [Q1] doi:10.1016/j.foodres.2013.09.007
53. Hassine, K. B., Taamalli, A., Malouche, D., Kammoun, N., Lazzez, A., Benincasa, C., ... Bouaziz, M. (2012). [Influence of variety, geographical site and extraction system on virgin olive oil \(VOO\) linoleic acid composition and its impact on consumer preference](#). *Linoleic Acids*, 97. [Q2]

Environmental science, natural products, agriculture and medical imaging

54. Naïja, A., Mohamed, O., Malouche, D., & Abiib, S. (2026). [Global phthalate occurrence and risk assessment in drinking water: a comprehensive Bayesian and probabilistic approach](#). *Science of the Total Environment*, 1051, 182291. [Q1] doi:10.1016/j.scitotenv.2026.182291
55. Losada-Echeberría, M., Naranjo, G., Malouche, D., Taamalli, A., Barrajón-Catalán, E., & Micol, V. (2023). [Influence of drying temperature and harvesting season on phenolic content and antioxidant and antiproliferative activities of olive \(*Olea europaea*\) leaf extracts](#). *International Journal of Molecular Sciences*, 24(1), 54. [Q1] doi:10.3390/ijms24010054

56. Mekki, I., Malouche, D., Smeti, S., Hajji, H., Mahouachi, M., Elloumi, M., & Atti, N. (2019). [Study of the breeding systems of sheeps in the montagnous area of north-western Tunisia](#). *Livestock Research for Rural Development*, 31(108). [Q3]
57. Khlaifi, H., Istrate, D., Demongeot, J., & Malouche, D. (2018). [Swallowing sound recognition at home using GMM](#). *IRBM*, 39(6), 407–412. [Q4] doi:10.1016/j.irbm.2018.10.009
58. Triki, H. Z., Laabir, M., Lafabrie, C., Malouche, D., Bancon-Montigny, C., Gonzalez, C., ... Daly-Yahia, O. K. (2017). [Do the levels of industrial pollutants influence the distribution and abundance of dinoflagellate cysts in the recently-deposited sediment of a Mediterranean coastal ecosystem?](#) *Science of the Total Environment*, 595, 380–392. [Q1] doi:10.1016/j.scitotenv.2017.03.183
59. Triki, H. Z., Daly-Yahia, O. K., Malouche, D., Komih, Y., Deidun, A., Brahim, M. et al. (2014). [Distribution of resting cysts of the potentially toxic dinoflagellate *Alexandrium pseudogonyaulax* in recently-deposited sediment within Bizerte Lagoon \(Mediterranean coast, Tunisia\)](#). *Marine Pollution Bulletin*, 84(1–2), 172–181. [Q1] doi:10.1016/j.marpolbul.2014.05.014
60. Ranjanomennahary, P., Ghalila, S. S., Malouche, D., Marchadier, A., Rachidi, M., Benhamou, C., & Chappard, C. (2011). [Comparison of radiograph-based texture analysis and bone mineral density with three-dimensional microarchitecture of trabecular bone](#). *Medical Physics*, 38(1), 420–428. [Q1] doi:10.1118/1.3528125

Conference proceedings

61. Rebhi, I., & Malouche, D. (2017b). [Decision making about products development through consumer preferences modeling based on descriptive characteristics of products](#). *IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA)*, 423–430. doi:10.1109/AICCSA.2017.202
62. Selmi, G., Azouz, Z. B., & Malouche, D. (2015). [The volume radius function: a new descriptor for the segmentation of volumetric medical images](#). *2015 International Conference on Image and Vision Computing New Zealand (IVCNZ)*, 1–6. doi:10.1109/IVCNZ.2015.7761572
63. Ghribi, K., Sevestre, S., Guessoum, Z., Gil-Quijano, J., Malouche, D., & Youssef, A. (2014). [A survey on multi-agent management approaches in the context of intelligent energy systems](#). *2014 International Conference on Electrical Sciences and Technologies in Maghreb (CISTEM)*, 1–8. doi:10.1109/CISTEM.2014.7077030
64. Aouinti, S., Mallek, H., Malouche, D., Saidi, O., Lassouedi, O., Hentati, F., & Romdhane, H. B. (2013). [Graphical interaction models to extract predictive risk factors of the cost of managing stroke in Tunisia](#). *2013 International Conference on Computer Medical Applications (ICCM)*, 1–6. doi:10.1109/ICCM.2013.6506162
65. Malouche, D., & Sevestre-Ghalila, S. (2008). [Estimating high dimensional faithful Gaussian graphical models by low-order conditioning](#). *Proceedings of the 26th IASTED International Multi-Conference on Applied Informatics, Artificial Intelligence and Applications*, 595–602.

Statistical methodology: graphical models, conditional independence, exponential families

66. Malouche, D. (2023). [Describing conditional independence statements using undirected graphs](#). *Axioms*, 12(12), 1109. [Q2]
67. Malouche, D. (2022). [Implications of faithfulness in graphical models](#). *Sankhya B: The Indian Journal of Statistics*, 84(2), 495–515. [Q4] doi:10.1007/s13571-021-00271-0
68. Malouche, D., & Rajaratnam, B. (2011). [Gaussian covariance faithful Markov trees](#). *Journal of Probability and Statistics*, 2011. [Q3] doi:10.1155/2011/152942
69. Malouche, D. (2009a). [Determining full conditional independence by low-order conditioning](#). *Bernoulli*, 15(4), 1179–1189. [Q1] doi:10.3150/09-BEJ193
70. Malouche, D. (2009b). [Mixed graphical model selection using Holm's procedure](#). *Communications in Statistics – Theory and Methods*, 38(9), 1453–1464. [Q3] doi:10.1080/03610920802455019
71. Kokonendji, C. C., & Malouche, D. (2008). [A property of count distributions in the Hinde–Demétrio family](#). *Communications in Statistics – Theory and Methods*, 37(12), 1823–1834. [Q3] doi:10.1080/03610920701809266
72. Letac, G., Malouche, D., & Maurer, S. (2002). [The real powers of the convolution of a negative binomial distribution and a Bernoulli distribution](#). *Proceedings of the American Mathematical Society*, 130(7), 2107–2114. [Q1] doi:10.1090/S0002-9939-02-05352-2
73. Letac, G., & Malouche, D. (2000). [The Markov chain associated to a Pick function](#). *Probability Theory and Related Fields*, 118(4), 439–454. [Q1] doi:10.1007/PL00008750
74. Malouche, D. (1998). [Natural exponential families associated to Pick functions](#). *Test*, 7(2), 391–412. [Q2] doi:10.1007/BF02565120

75. Malouche, D. (1997). [L'action quadratique du groupe des homographies sur les familles exponentielles réelles](#). *Comptes Rendus de l'Académie des Sciences – Series I – Mathematics*, 325(9), 1029–1032. [Q3] doi:[10.1016/S0764-4442\(97\)89099-4](#)